

Assignment 4: Horizons and Congruences

to be handed in online by Sunday, June 7th

Students taking the “[...] Black Holes” physics course are only expected to submit the exercises marked with an asterisk*, but are obviously invited to try all of them. Students taking the “[...] Classical Systems” maths course are expected to submit (almost) all exercises.

If you skip an exercise, be sure to read it at least, since you might need its result later.

Feel free to use **Mathematica**, but submit the notebook or screenshots of it.

Exercise 1: Eddington-Finkelstein Coordinates

A general static, spherically symmetric metric can be written as

$$ds^2 = -A(r) dt^2 + \frac{dr^2}{B(r)} + r^2 d\Omega^2, \quad (1)$$

where $d\Omega^2$ is the metric on a unit 2-sphere. Assume that $A(r)$ and $B(r)$ are smooth functions of r such that both have a simple zero at $r = r_+ > 0$ and are positive for $r > r_+$.

a)* Show that radial null geodesics are given by $t \pm r_* = \text{constant}$, where

$$r_* = \int_{r_0}^r \frac{dx}{\sqrt{A(x)B(x)}}, \quad (2)$$

with $r_0 > r_+$ an arbitrary constant. Show that $r_* \rightarrow -\infty$ as $r \rightarrow r_+$.

b)* Obtain the metric in ingoing Eddington-Finkelstein coordinates (v, r, θ, ϕ) , with $v = t + r_*$. Explain why this metric can be analytically continued through $r = r_+$.

Exercise 2: Acceleration of Killing Orbits

In a spacetime that possesses a timelike Killing vector field k , consider trajectories of massive particles that move along orbits of k , that is, their four-velocity is a unit-normalized timelike vector u that is proportional to k .

a)* Show that the acceleration of these particles, $a^\mu = u^\nu \nabla_\nu u^\mu$, can also be written as

$$a_\mu = \nabla_\mu \log \sqrt{-k_\nu k^\nu}. \quad (3)$$

Can you now identify a function $\phi(x)$ that can be interpreted as the gravitational potential in which the particle moves?

b)* Compute a^μ for particles moving along orbits of $k = \partial_t$ in the Schwarzschild metric

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2 d\Omega^2. \quad (4)$$

What kind of situation do these trajectories represent? Obtain the gravitational potential ϕ . Can you interpret these results?

c)* Evaluate a^μ as $r \rightarrow 2M$. Can you identify this result? Read Harvey's 9.2.

Exercise 3: Surface Gravity Identities

Let $\mathcal{H}$ be a Killing horizon with KVF ξ^μ . We will assume that ξ^μ is timelike on one side of $\mathcal{H}$ (the “exterior” of the black hole, connected to asymptotic infinity). Let us also define $|\xi| := \sqrt{-\xi_\mu \xi^\mu}$ and the surface gravity of the Killing horizon $\mathcal{H}$ as the non-affinity of ξ^μ ,

$$\xi^\mu \nabla_\mu \xi^\nu =: \kappa \xi^\nu. \quad (5)$$

a) Use Frobenius' Theorem to prove

$$\kappa^2 = -\frac{1}{2} (\nabla^\mu \xi^\nu) (\nabla_\mu \xi_\nu) |_{\mathcal{H}}. \quad (6)$$

b) Prove that $\xi^\mu \nabla_\mu |\xi| = 0$.

c) Defining a as the acceleration of ξ -orbits, as in exercise 2, prove

$$\kappa^2 = \lim_{\rightarrow \mathcal{H}} |\xi|^2 a^2 = \lim_{\rightarrow \mathcal{H}} \frac{(\xi^\mu \nabla_\mu \xi^\nu)(\xi^\rho \nabla_\rho \xi_\nu)}{|\xi|^2}. \quad (7)$$

Exercise 4: Traversable Wormholes

The geometry

$$ds^2 = -dt^2 + d\rho^2 + (\rho^2 + a^2) (d\theta^2 + \sin^2 \theta d\phi^2), \quad (8)$$

where a is a real constant $-\infty < \rho < +\infty$, is a simple model of a traversable wormhole. You should convince yourself of this interpretation.

Consider a congruence of radial null geodesics in this geometry, with $\xi = \partial_t - \partial_\rho$ as its generator.

a)* Argue that this vector describes null rays that are ingoing in the region $\rho > 0$. Are they also ingoing in the region $\rho < 0$?

b)* Argue that the vorticity of this congruence is zero. Conclude that the shear vanishes as well. The more efficiently you obtain these results, the better.

c)* Compute the expansion θ of the congruence (again, be efficient). Interpret the result: is the congruence always converging? Then compute the rate of change of the expansion $\xi^\mu \partial_\mu \theta$.

d)* Insert the previous results into Raychaudhuri's equation to obtain the null-null component $\xi^\mu \xi^\nu T_{\mu\nu}$ of the stress-energy tensor that would be required in order that this geometry solves the Einstein Equations. What can you say about energy conditions in this spacetime?